

Experiment 9: Map Coloring (CSP)

Aim

To solve the Map Coloring Problem using CSP.

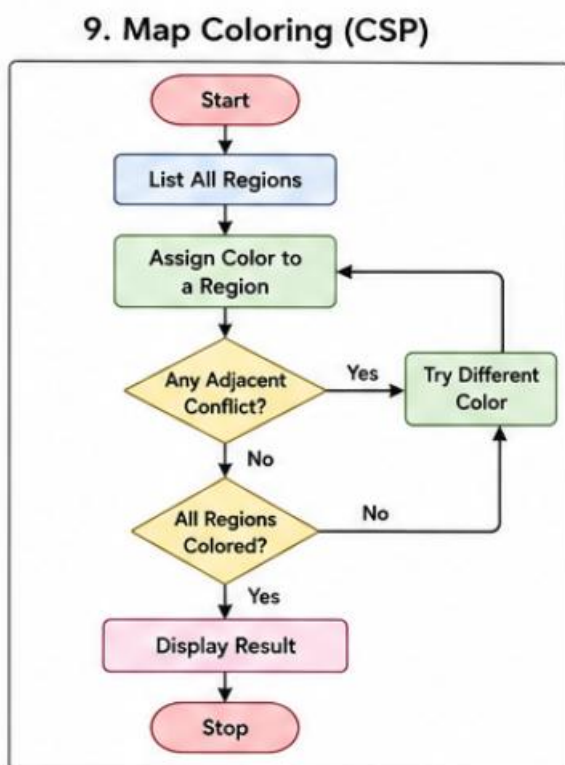
Objective

To assign different colors to neighbouring regions.

Algorithm

1. Define regions.
2. Assign colors.
3. Check neighbouring regions.
4. Ensure adjacent regions have different colors.
5. Display the solution.

Flowchart



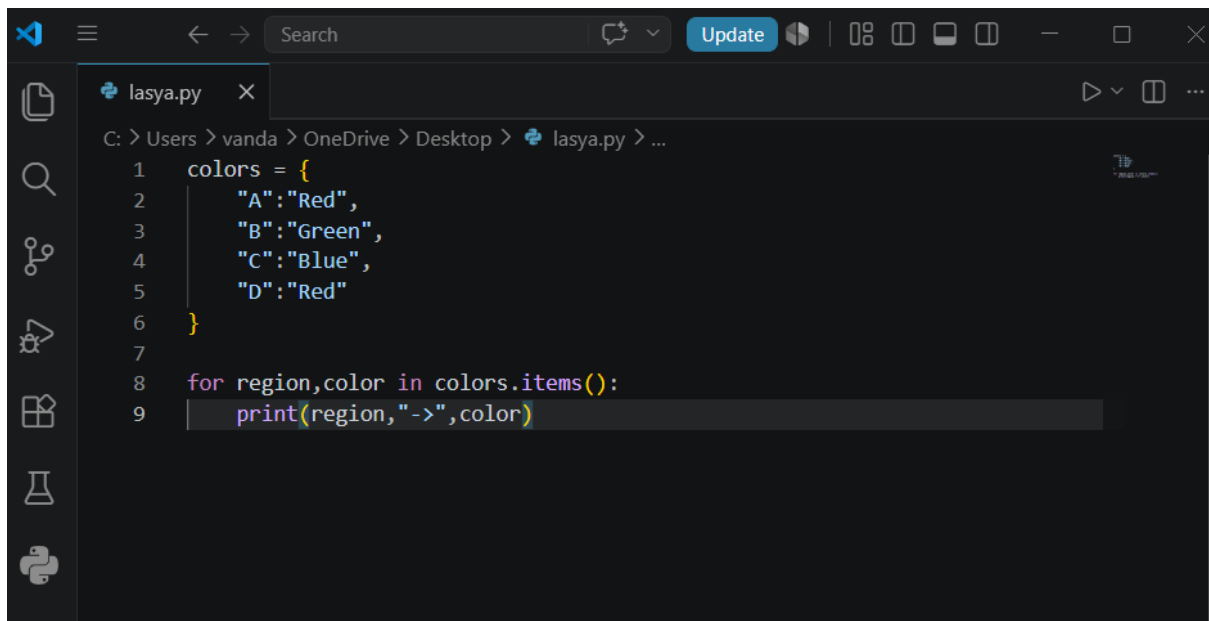
Code

```
colors = {
```

```
"A":"Red",  
"B":"Green",  
"C":"Blue",  
"D":"Red"  
}
```

```
for region,color in colors.items():  
    print(region,"->",color)
```

Output :



```
lasya.py  
C: > Users > vanda > OneDrive > Desktop > lasya.py > ...  
1  colors = {  
2      "A":"Red",  
3      "B":"Green",  
4      "C":"Blue",  
5      "D":"Red"  
6  }  
7  
8  for region,color in colors.items():  
9      print(region,"->",color)
```

```
PS C:\Users\vanda\onedrive\desktop> python lasya.py  
A -> Red  
B -> Green  
C -> Blue  
D -> Red  
PS C:\Users\vanda\onedrive\desktop> █
```

Result

The regions were colored successfully.

Conclusion

The CSP approach assigns colors without violating constraints.